

The amplitude writing as a single-parameter probe

The production law and the amplitude writing, both with the exponent pinned at $p = 1$ and the law origin pinned at the nominal contact $\bar{w}_s = 1$:

$$\text{Form B (length, production)} \quad \bar{a} = 1 - \frac{b}{(\bar{w} - 1) + b}$$

$$\text{Form C (amplitude, this probe)} \quad \bar{a} = 1 - \frac{b}{\bar{w}}$$

Form C is Form B with the two states tied. Setting $\bar{w}_s = b$ in the production law with $p = 1$,

$$1 - \left(1 + \frac{\bar{w} - b}{b}\right)^{-1} = 1 - \frac{b}{\bar{w}},$$

identically. The probe therefore needs no new law: it opens a single free scalar β along the $(b, \bar{w}_s)$ diagonal of the existing controller. By the chain rule along that diagonal, evaluated at the tied point,

$$J_\beta = J_b + J_{w_s} = \frac{1}{\bar{w}^2}, \quad \frac{\partial \bar{a}}{\partial \beta} = \frac{\partial \bar{a}}{\partial b} + \frac{\partial \bar{a}}{\partial \bar{w}_s} = -\frac{1}{\bar{w}},$$

both exact. These are the closed forms already carried by `formB_amp_functions.tex`.

Why the probe exists. The production Jacobian $J_b = (g - b)/(g + b)^3$, $g = \bar{w} - 1$, *vanishes at* $g = b$, i.e. $\bar{w} = 1 + \hat{b} \approx 2.125$ ($h = 4.78 \mu\text{m}$). The canonical oscillation spans $h \in [4.5, 9.5] \mu\text{m}$, so the production arm crosses its own b -channel null twice per cycle and sits $0.28 \mu\text{m}$ from it during the final hold, where $|J_{w_s}/J_b| = 2b/|g - b| = 18$. The tied Jacobian $1/\bar{w}^2$ is strictly positive with no null anywhere. The probe asks whether that difference is what has kept $\hat{b}$ from moving.

Scope limitation against the SSOT. `formB_ws_ref.tex` S1 rules the amplitude writing out on two grounds: fitted with a free origin it returns $\bar{w}_s = -0.063$, not a surface position; and it drags fractional powers and logarithms into every Jacobian. Neither applies here — the origin is pinned at 1 and the exponent at 1, so the Jacobians above are rational and the origin is not a fitted quantity. The SSOT stands; this file records only the scoped exception.

What the tie costs, stated before the run. At the same b the production law is evaluated at $g = \bar{w} - 1$ and the tied law at $\bar{w}$: these are two different model classes, not a reparametrisation of one filter. On the canonical envelope the anchored-seed shape floor is 0.00284 for Form B against 0.03724 for Form C, a factor 13.1; at the final hold ($\bar{w} = 2$) the anchored gains are -0.01% and -7.04% against the truth. A worse gain error in the amplitude arm is therefore the *expected* outcome and is not evidence against the probe; the criteria are the b -channel ones.

The prior width, by the six-step chain

Steps follow `reference/shared/param_prior_rules.md`.

(2) Seed from an analytic anchor. The far-field truth deficit is $1 - \bar{a} \simeq (9/8)/\bar{w}$, and the amplitude writing's deficit is $b\bar{w}^{-p}$, so *one* limit pins both constants: $b_0 = 9/8$, $p_0 = 1$. The ignorance source of the error is zero.

(3) The reading. Taking logarithms of the deficit,

$$\ln(1 - \bar{a}) = \ln b - p \ln \bar{w},$$

so b is the log-log *intercept* and p is the slope. With p pinned the intercept is read directly against the truth — unlike the production writing, where b sits inside a bracket and the level-versus-slope choice needs the argument of S10:

$$b_{\text{eff,C}}(\bar{w}) = \bar{w} (1 - \bar{a}_{\text{true}}) = \bar{w} \frac{c-1}{c}, \quad b_{\text{eff,B}}(\bar{w}) = (c-1)(\bar{w}-1).$$

Both run from 1 at contact to 9/8 in the far field: each traverses exactly the 12.5 % anchor tension, and the global sup *is* that tension (0.1250 and 0.1244; the difference is the sweep starting at $\bar{w} = 1.001$).

(5)–(6) The sup domain and the width. On the canonical envelope [1.90, 22.3] (commanded trough 2.0 less the derived margin 0.1; start height plus 1),

$$\sqrt{P_{\beta\beta}[0]} = \sup_{\text{envelope}} |b_{\text{eff,C}} - 9/8| = 0.0708, \quad (\text{production: } 0.0157).$$

Two independent criteria select the envelope over the global fallback 0.125. Rule 3 (*just enough, not more*): the value the truth demands at the final hold is $b_{\text{eff,C}}(2) = 1.0588$, a travel of 0.94σ against 0.0708 but only 0.53σ against 0.125 — $1.9\times$ over-wide, the regime the 2026-08-01 revision exists to remove. And the $P[0]$ budget: at the predicted $\sim 90\%$ contraction the realised ratio $(\Delta\beta)^2/(P[0] - P[\infty])$ is 0.97 against a limit of 1, versus 0.31 for the global width. The envelope prior is TIGHT by construction, i.e. $\sup |\theta_{\text{eff}} - \theta_0| \div \sqrt{P[0]} = 1.00\times$, the same honesty status the production widths carry.

The shape floor on the same envelope, which funds $P_{\bar{a}\bar{a}}[0]$, is

$$\text{floor}_{a,C} = \sup_{\text{envelope}} \left| 1 - \frac{9/8}{\bar{w}} - \frac{1}{c} \right| = 0.03724 \equiv \sup_{\text{envelope}} \frac{|b_{\text{eff,C}} - 9/8|}{\bar{w}},$$

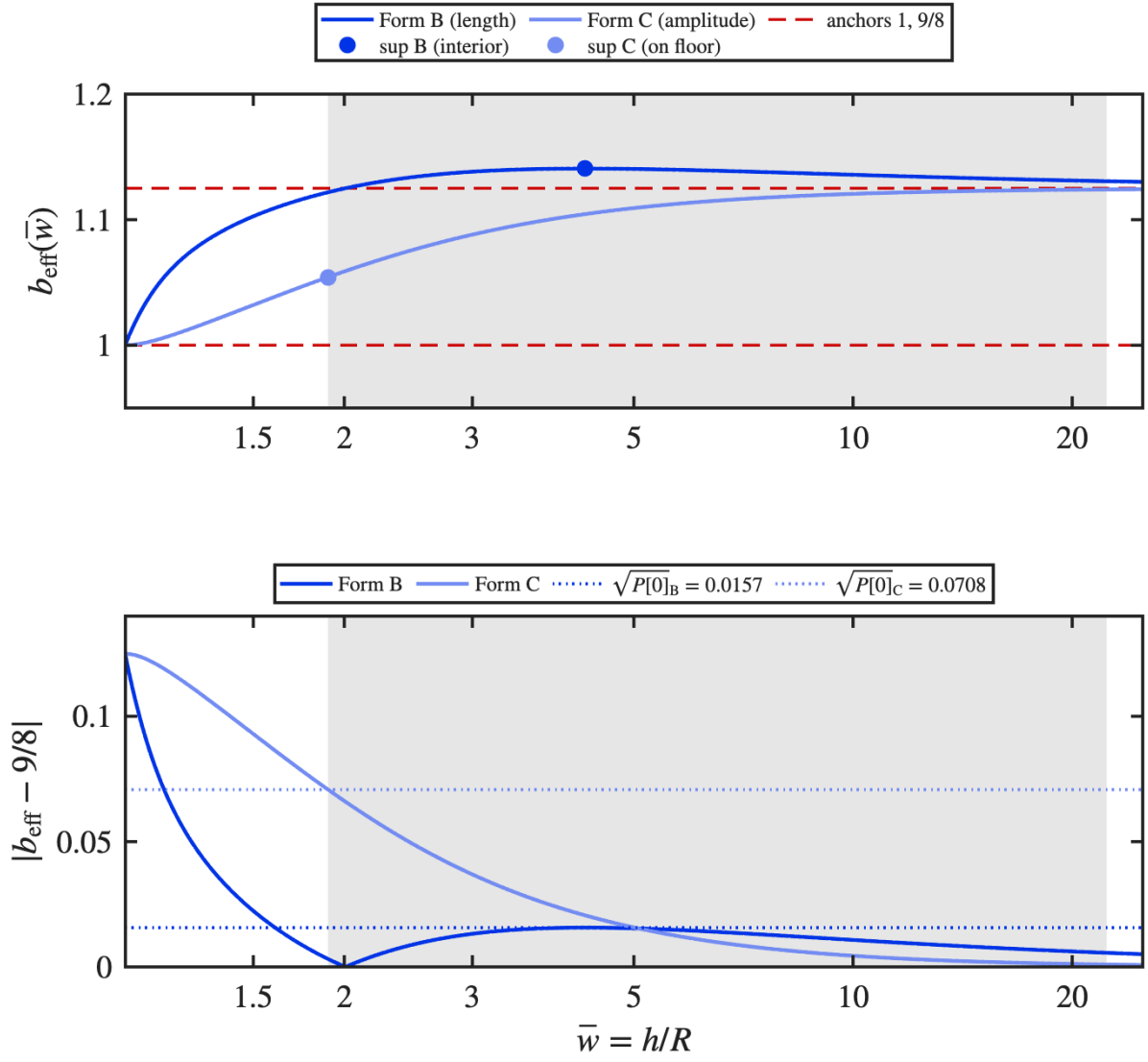
the identity holding pointwise and used as the driver’s self-check (residual 1.4×10^{-17}).

Declared boundary sensitivity

$b_{\text{eff,C}}$ is strictly monotone, so its sup sits on the envelope *floor*; $b_{\text{eff,B}}$ is not, and its sup sits at the interior point $\bar{w} \approx 4.28$. Refitting the floor:

envelope floor	1.90	1.95	2.00
$\sqrt{P[0]}$, Form B (production)	0.0157	0.0157	0.0157
$\sqrt{P[0]}$, Form C (this probe)	0.0708	0.0685	0.0663

The production comment that the envelope margin is “insensitive by design” is therefore true only for the length writing. For the amplitude writing the margin’s provenance is load-bearing and is restated here: 0.1 is the measured true-trough undershoot $0.033 R$ plus position measurement noise $\sim 0.05 R$, rounded up. A $0.05 R$ error in that margin moves the declared prior by 6.4 %. This is a declared sensitivity, not a tuning knob.



Step 3 of the $P[0]$ chain for both writings, with nothing fitted: each curve is the constant its own law demands of the published truth at each height. Top — both readings travel from the contact anchor 1 to the far-field anchor $9/8$, so the global sup is the 12.5% anchor tension in either writing; the shaded band is the canonical envelope $[1.90, 22.3]$ and the filled circles mark where each sup is attained. Bottom — the same quantity as a prior budget. The production reading is non-monotone, crosses its own anchor near $\bar{w} \approx 2.5$ and peaks in the interior, which is why its width 0.0157 is identical for envelope floors 1.90/1.95/2.00. The amplitude reading is monotone, so its width 0.0708 is attained on the floor itself and inherits the margin's provenance directly (0.0708/0.0685/0.0663 across the same three floors, a 6.4% spread). The $4.5\times$ ratio between the two widths is not a modelling choice — it is what the two writings honestly cost on the heights this scenario visits, and it is half of why the amplitude arm's b channel is expected to be live where the production one is not (the other half being the Jacobian null at $\bar{w} = 1 + \hat{b}$, which the amplitude writing does not have).

Result: the arm ladder

Plant = the real plane wall; the controller knows nothing about it and every seed is the far-field anchor. Twelve seeds, one scenario, identical noise realisation per seed across arms. Each rung changes exactly one thing, so the three effects add. Structural acceptance (`verify_formB_amp_tie`: default bit-identity, the closed-form fold to 9×10^{-16} , the tie verified inside the shipped controller to 1.3×10^{-15} , finite differences, the pole guard, covariance health) passed before any number here was read.

	arm	$\hat{\beta}$ final	$\sqrt{P}$ shrink	budget	desc %	hold %
A0	<code>len_prod</code> (b,p free)	1.1247 ± 0.0035	2.1%	1.15	1.55	-0.39
A1	<code>len_bonly</code>	1.1246 ± 0.0054	3.2%	1.74	1.10	-0.47
A2	<code>len_wide</code> (prior $\times 4.5$)	1.1207 ± 0.0500	35.0%	0.81	2.10	-0.30
A3	<code>len_ws98</code> (origin 9/8)	1.0923 ± 0.0312	14.6%	1.47	6.93	-4.98
A4	<code>amp</code> (the tie)	1.0821 ± 0.0340	38.1%	1.00	6.60	-3.08

Travel decomposition (mean over 12 seeds, units of R):

A1 <code>len_bonly</code> , production prior	-0.00037	
+ prior width $0.0157 \rightarrow 0.0708$	-0.00396	(A1 $\rightarrow$ A2)
+ law origin $1 \rightarrow 9/8$	-0.02834	(A2 $\rightarrow$ A3)
+ Jacobian fold $J_b \rightarrow J_b + J_{w_s}$	-0.01020	(A3 $\rightarrow$ A4)
= A4 <code>amp</code>	-0.04287	

Three readings, in order of how much they change the mainline.

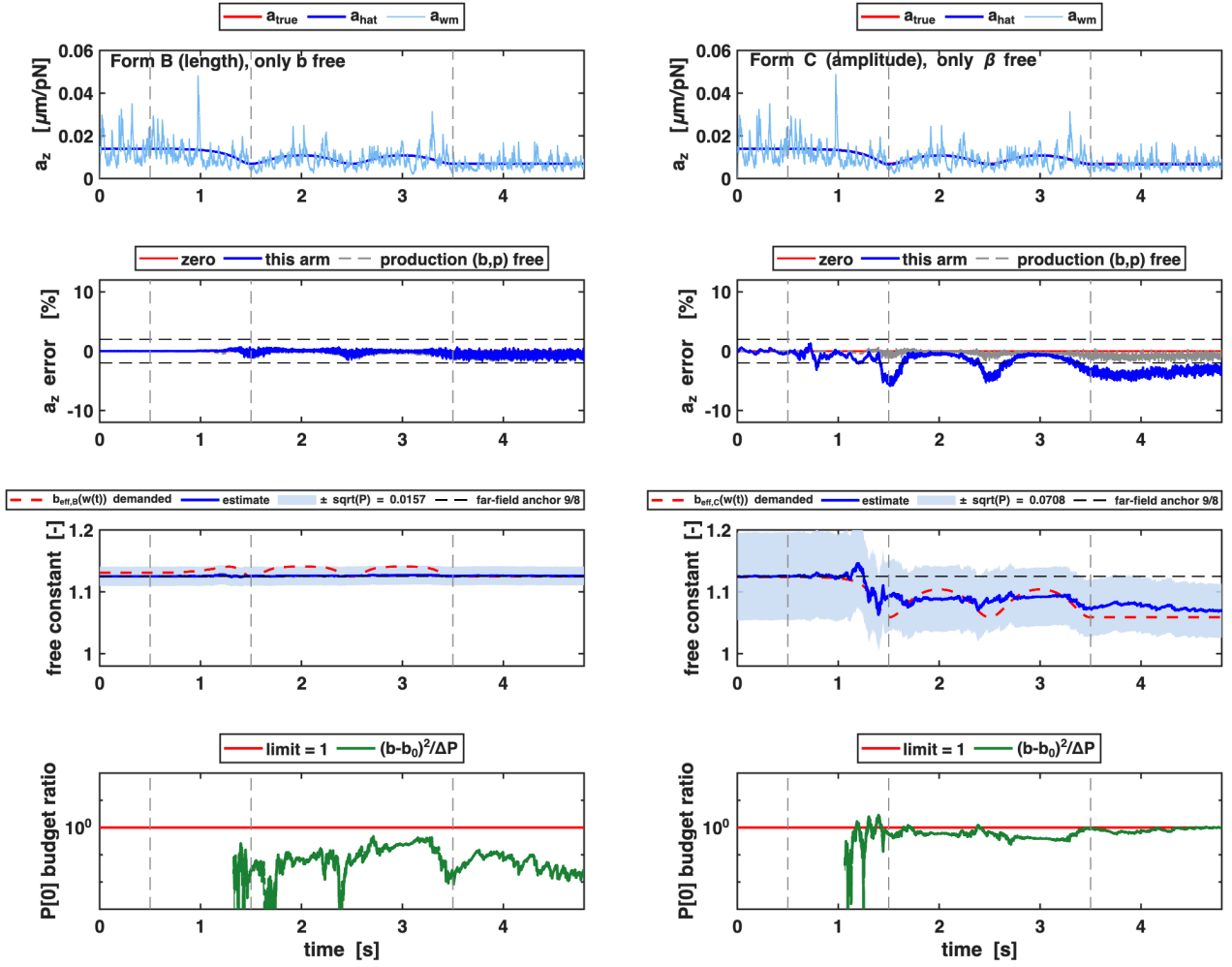
(i) Widening the prior alone buys essentially nothing. A2 opens the b prior by $4.5\times$ and $\hat{b}$ still does not move (-0.06σ , 9% of the total travel) — yet its variance drops 35%, almost as much as the tied arm’s 38%. *Variance reduction and travel are different things.* The length writing’s b channel does carry information; it has nothing to say, because on this envelope the anchored b is already what the truth demands to within 0.0157. On the 2026-08-06 arm the truth was a constructed curve 63.5 widths away and the prior was the binding constraint; with the real plane wall it is not.

(ii) The Jacobian fold is measurable and one-directional. A3 and A4 share the law at $t = 0$, the origin, the prior and the seed noise; they differ only by $J_b \rightarrow J_b + J_{w_s}$. The fold adds 33% more travel (-0.0429 vs -0.0312), $2.6\times$ more variance reduction (38.1% vs 14.6%), and cuts the residual hold error from -4.98% to -3.08% . Along the real trajectory the tied Jacobian is $+0.25$ throughout the final hold where the production one is -0.013 — a factor 19, at the height where the gain matters most and where the production J_b has changed sign.

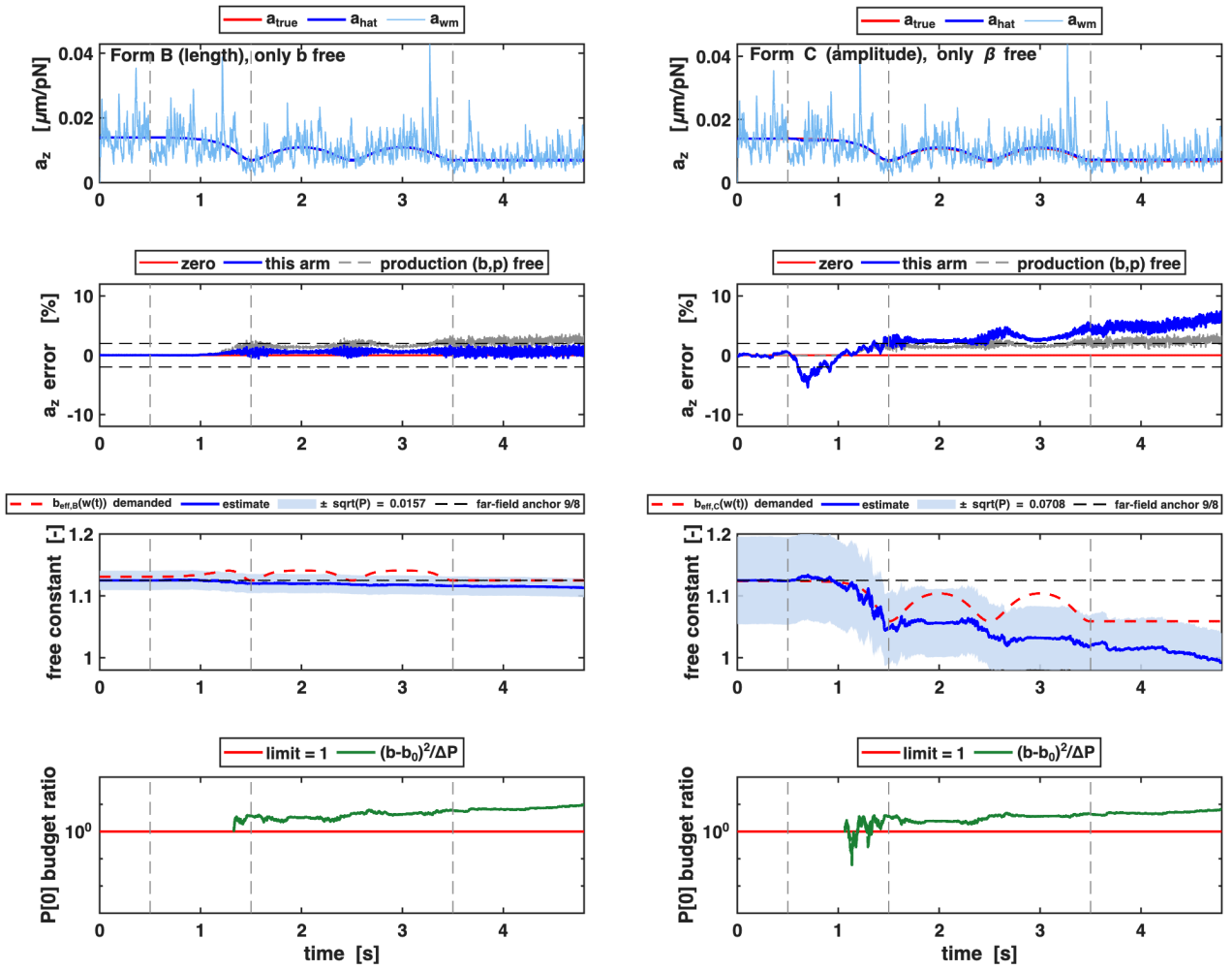
(iii) Where $\hat{\beta}$ lands was predicted correctly. Mean 1.0821 against the pre-registered information-weighted $\langle b_{\text{eff}} \rangle = 1.0833$; $t = -4.36$ (df 11) against the “did not move” hypothesis. The estimate does *not* reach the envelope minimax 1.0651 or the trough-exact 1.0588, exactly as the derivation predicted: the prior width is set by the trough, which is the least informative point on the trajectory, while the information sits in the high-velocity descent. Consistency: $\text{corr}(\hat{\beta}, \text{hold error}) = -0.80$, the sign the law requires; the traversal budget averages 1.003 against a limit of 1, i.e. the arm consumes exactly the prior it declared. Honesty: spread over seeds $\div$ claimed final $\sigma = 0.78$. One genuine outlier, seed 11 ($\hat{\beta} = 0.992$, hold $+5.3\%$, the only positive one); it is an outlier in A3 as well, so it belongs to that noise realisation and not to the writing. Excluding it the mean is 1.0903 with spread 0.0199.

What this does not license. As a candidate production law the amplitude writing fails on accuracy, as stated before the run: descent 6.60 % and hold -3.08 % against production's 1.55 % and -0.39 %. Most of its travel is repairing a 8.4 % seed model error it inflicted on itself by moving the law's zero-gain point to $\bar{w} = \beta$. Its final absolute uncertainty is also worse ($\sqrt{P} = 0.0438$ vs the length arm's 0.0152), because the length writing's prior is tighter — and it is tighter because that writing genuinely fits the truth better on the heights this scenario visits. The `formB_ws_ref` S1 ruling stands.

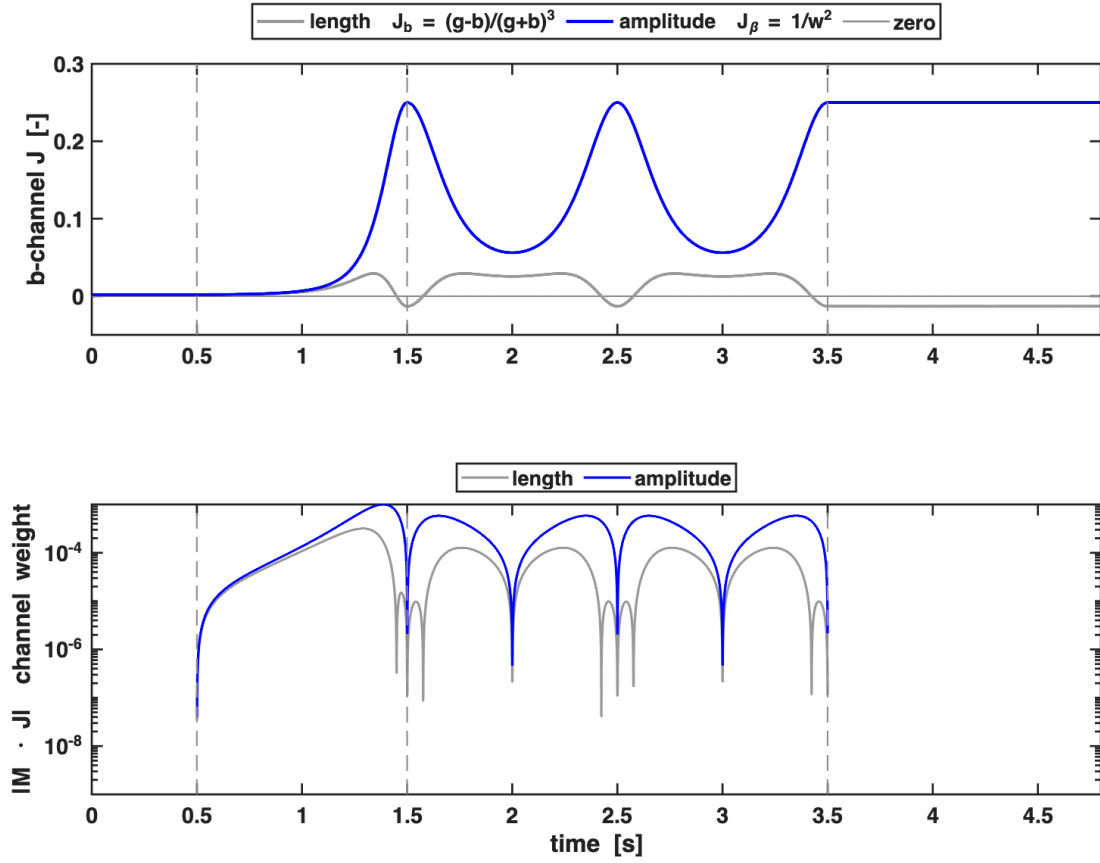
What it does settle. The production arm's $\hat{b}$ standing still is *correct behaviour, not a defect*: A2 shows that releasing the prior does not move it, and A4 shows that the same filter moves $\hat{\beta}$ by -0.61σ reproducibly once there is something to learn. The b-channel weakness diagnosed from the Jacobian null is real and worth 33 % of the travel, but it is not what has been holding the mainline back.



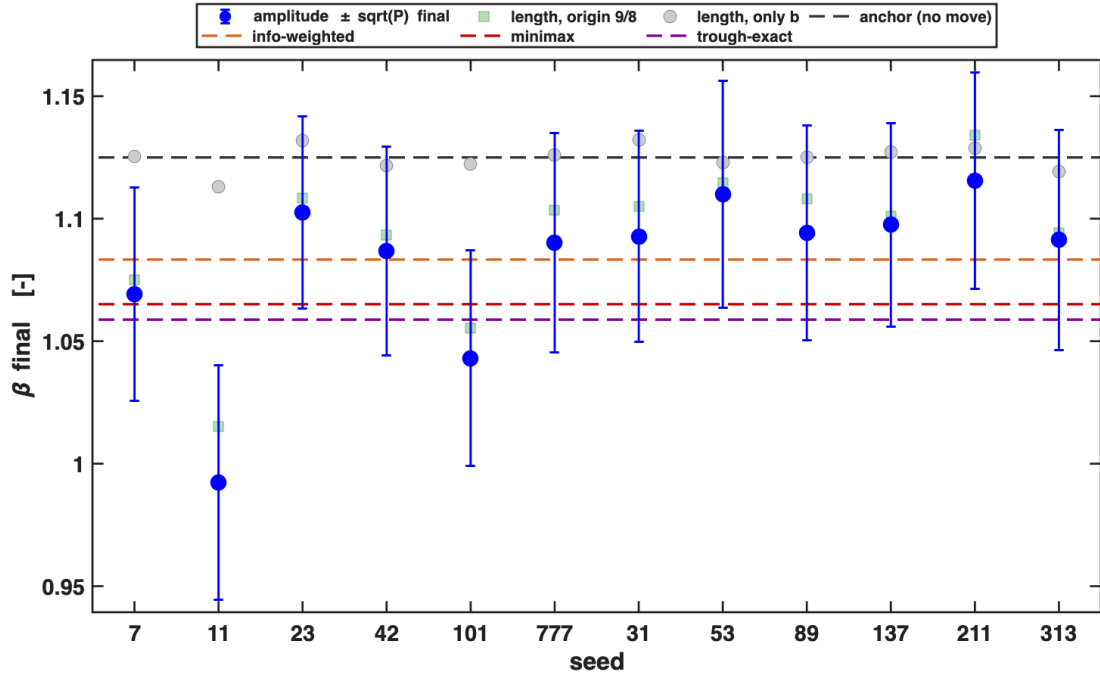
Seed 7, both writings, only the shape constant free. Left the production length writing, right the tie. Row 3 is the whole story: the length arm's estimate sits on the anchor and the red dashed curve — the constant the truth demands at the height the particle is at — sits on top of it, so there is nothing to correct. The amplitude arm starts on the same anchor, walks down through the descent and then tracks its own demanded curve. Row 4 shows the traversal consuming the declared prior (ratio $\rightarrow 1$) rather than exceeding it. Row 2 is the price: the amplitude arm runs at -3 to -6% where the production reference (grey) sits inside $\pm 1\%$. Note also that the amplitude arm's error is largest in the oscillation troughs and recovers at the peaks — the signature of a one-parameter family whose single constant cannot be right at two heights at once.



Seed 11, the outlier, shown rather than dropped. Here $\hat{\beta}$ overshoots to 0.992 — past every pre-registered target — and the hold error goes +5.3 %, the only positive one in the set. The same seed is the extreme of the A3 arm too (1.0151), so this is a property of that noise realisation and not of the writing: a run whose descent innovations happen to push the constant too far, with nothing downstream able to pull it back because $Q_{\beta\beta} = 0$ means the filter's confidence only ever grows. This is the failure mode a single free constant with a hard zero process noise has, and it is the reason the spread over seeds — not the single-run number — is the reproducibility statistic.



The b -channel Jacobian along the real trajectory (seed 7, evaluated at the estimate the controller actually used). Top — the production J_b (grey) sits near zero, dips *negative* at every oscillation trough and stays negative through the entire final hold; the tied $J_\beta = 1/\bar{w}^2$ (blue) peaks at 0.25 at each trough and holds there, a factor 19 at the height where the gain matters most. Five sign changes after the initial hold. Bottom — the same quantity weighted by the commanded motion it enters with, $|M \cdot J|$, on a log axis: the grey trace carries *extra* nulls that the blue one does not, one pair per cycle, and those are the Jacobian zeros, distinct from the trajectory turning points where both traces null. This is the offline page-5 prediction realised in closed loop.



Final constant per seed against the four pre-registered landing hypotheses. Grey circles: the length writing with only b free — pinned on the anchor, spread 0.0054, nothing learned. Green squares: the same writing with its origin declared at 9/8 (the fold negative control). Blue with error bars: the tie, showing its own claimed final $\sqrt{P}$. The blue mean 1.0821 lands on the information-weighted prediction 1.0833 and clears the “did not move” line at $t = -4.36$; it does not reach the minimax or trough-exact lines, which is what the derivation said would happen because the prior is sized by the trough while the information sits in the descent. Eleven of twelve seeds move monotonically downward; seed 11 overshoots.